Problem and user definition

AfyaPlus AI

Triage support and supply monitoring for clinics with limited resources

Design aim: fast assistance within privacy and operating-cost limits.

PRIMARY USER

Clinical officers

At the point of care

Quick triage guidance alongside supply availability.

Scope: decision support under clinical oversight.

Chosen engineering path

Official AfyaPlus Module 4 Capstone

One integrated capstone path

FastAPI triage boundary with logistics Model Context Protocol (MCP) tooling.

The README declares the official path and reuses Weeks 1–10 frameworks.

Weeks 1–5

Data containment and routing stubs

Weeks 6–7

API boundary and cost model

Weeks 8–9

Model pins and masking checks

Week 10

Data-transfer policies

System infrastructure architecture

Declared design: a boundary layer, a pre-model guard and a privacy-aware log tracker

01 API boundary

FastAPI request routing
Authorization context
Role and symptom input

→

02 Injection guard

Screen before model use
Reject unsafe input
HTTP 400 demo target

→

03 Log tracker

Remove sensitive strings
Hash payload with SHA-256
Write tracking record

Triage router and local MCP stub sit within the application pipeline.

Week 8 model pins (integration_map.json)
openai/gpt-4o-mini@2026-10-11-snapshot
sokonihub/lab-bow-v1@a4f1082c9e7b238

Schematic only. The supplied excerpts do not establish exact call order or deployment behavior.

Week resource skill coverage

Repository navigation mapped to the declared curriculum reuse

Week	Capability	Source artifact
1–5	Data containment and routing stubs	README.md
6	API endpoint routing	main_pipeline.py
7	FinOps cost optimization	docs/cost_analysis.md
8	Model signature version pins	integration_map.json
9	Log masking smoke checks	tests/test_security_smoke.py
10	Operational data-transfer policies	README.md

Live application demo

Two-stage presenter runbook. The triage route still needs confirmation.

1 Start the local service

```
python -m uvicorn main_pipeline:app \
  --host 127.0.0.1 --port 8000
```

2 Send an authorized request

```
curl -X POST "$TRIAGE_URL" \
  -H "Authorization: Bearer test_token" \
  -H "Content-Type: application/json" \
  -d '{"role":"clinical_officer",
       "symptoms":"Patient shows a high fever."}'
```

Set TRIAGE_URL to the confirmed local endpoint. The pasted README curl URLs are incomplete.

DEMO CHECKS

Authorized input

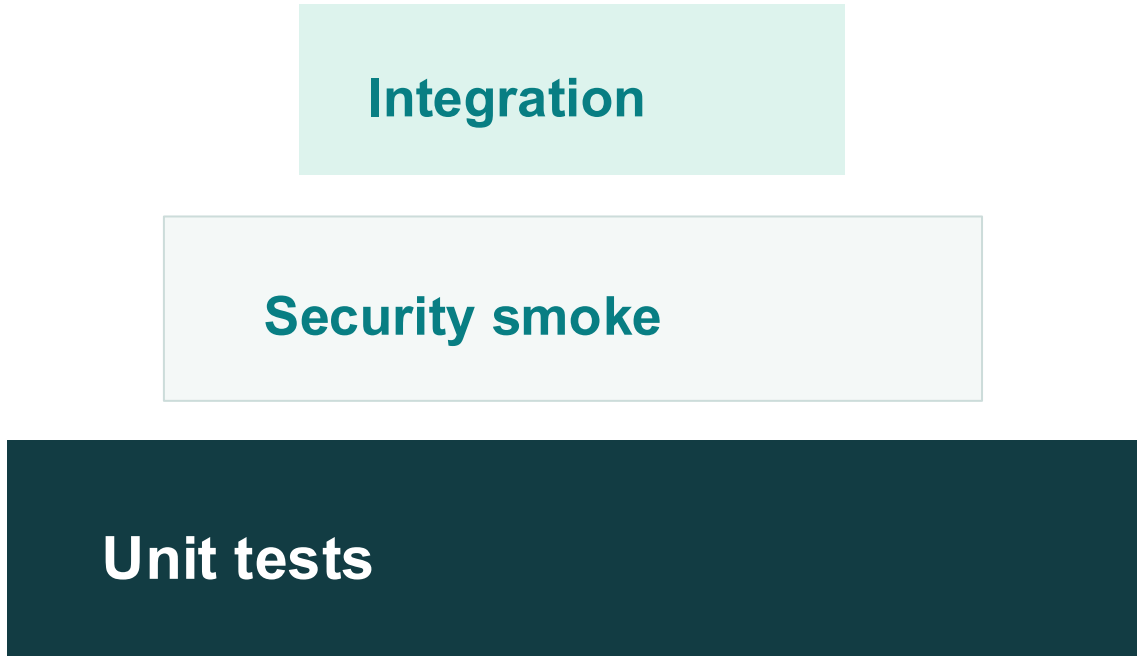
Inspect the triage response.

Injection fixture

Expected HTTP 400
Outcome not yet demonstrated here.

Testing pyramid evidence

Local benchmark: execute_triage_router_stub on CPU-Standard-Lab-Fallback




```
python -m pytest -q
```

Assertions in the supplied code

Triage case: RED_ALERT and EMERGENCY_TRANSFER
Model pin: triage entry contains “@”
Evaluation function: score == 1.00

300 synthetic patient entries

README fixture count. A 1.00 assertion does not establish clinical precision.

p95: 0.0542 ms Average: 0.0451 ms
100 test cycles. Source: perf_stub.json
Stub timing only, not end-to-end latency.

Security and compliance engineering

Logging incident: 12 October 2026. Legal review remains open.

Incident and reported resolution

Fault	Leading whitespace bypassed token checks, exposing raw authorization strings in logs.
Fix	The note reports string removal and SHA-256 payload hashing.
Regression check	tests/test_security_smoke.py Credential redaction validations. Pass reported in the incident note.

Privacy controls

main_pipeline.py

Input validation filters, token masking, retention loops and restricted MCP access.

fixtures/triage_messages_sample.jsonl
300-row testing dataset

Evidence still needed

Asset inventory, retention schedule and sweeper run records.

KDPA alignment is an aim. Lawful-basis review remains pending.

Financial projections and checkpoints

Source: docs/cost_analysis.md. Monthly USD forecast, reported 11 October 2026 rates.

Monthly scenario	Baseline	10× surge
Requests	100,000	1,000,000
Token processing	$17.90	$179.01
Infrastructure floor	$520.00	$800.00
Total monthly cost	$537.90	$979.01
Cost per 1,000 requests	$5.3790	$0.9790

Optimization levers

FAQ TTL cache: 62.5% hit rate reported
Offline batch queue for logistics updates

Financial Credit Policy Exception Notes

The AI path serves only as an informational recommendation advisor to clinical staff.
It recommends rather than decides on credit or clinical asset distribution.
Human management teams retain all final financial authorizations.

IMPLEMENTATION STATUS

Reported implemented

FastAPI triage layer
Logging fix and smoke checks

Declared fallbacks

Local MCP stub
Local-env plus twin JSON
No Azure Key Vault or AWS

Open checkpoints

Lawful-basis review, demo and run evidence